Weekly Report 3: Dummy Silhouette and Voxel-Based 3D Sphere Reconstruction

Project Number: 18

1 Introduction

Building upon our previous work on voxel grid creation and visualization, this week we implemented the next major step toward 3D object reconstruction — **visual hull carving using silhouette images**. The main goal was to verify that our voxel grid structure can be used to reconstruct simple 3D models from multiple 2D silhouettes. To validate this, we generated a dummy circular silhouette and successfully reconstructed a **dummy 3D sphere model** from it, thus confirming that our voxel carving logic works as intended. Our next step will involve replacing the dummy silhouette with the real diamond stone silhouette images for actual reconstruction.

2 Methodology

We extended our voxel grid pipeline with silhouette generation and carving logic. The core idea is to start with a uniform voxel grid representing 3D space and iteratively remove ("carve") voxels that do not project inside silhouettes from multiple viewing angles.

2.1 Voxel Grid Creation

A voxel grid of resolution $64 \times 64 \times 64$ was created using `NumPy`. Each voxel corresponds to a 3D coordinate point within a normalized cube (-1 to 1 range). This grid forms the foundation upon which silhouette carving is applied.

2.2 Dummy Silhouette Generation

To simulate multiple 2D views, we created synthetic silhouettes representing a rotating circle (projected sphere). Each silhouette mask was generated mathematically using the circle equation:

$$x^2 + y^2 < r^2$$

where $r = 0.5$ defines the radius of the silhouette region.

2.3 Visual Hull Carving

For each generated silhouette, the voxel grid was rotated by a specific viewing angle and projected into 2D. Voxels that fell outside the silhouette boundary were carved away. This process was repeated across multiple ($n = 20$) simulated viewpoints to create a refined 3D volume approximating the original shape.

Implementation Code Snippet:

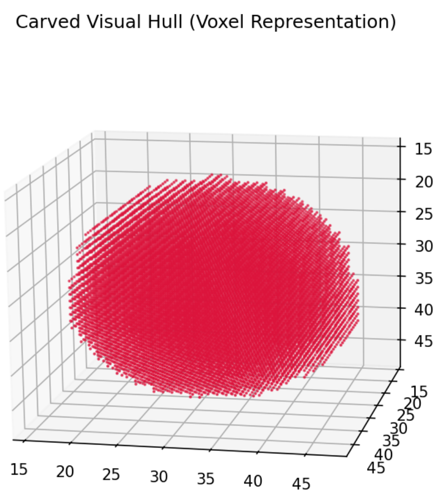
```
def carve_visual_hull(voxel_grid, n_views=20):
    res = voxel_grid.shape[0]
    mask = np.ones((res, res, res), dtype=bool)

    for i in range(n_views):
        angle = np.deg2rad(i * (180 / n_views))
        R = np.array([
            [np.cos(angle), 0, np.sin(angle)],
            [0, 1, 0],
            [-np.sin(angle), 0, np.cos(angle)]
        ])
        rotated = voxel_grid.reshape(-1, 3) @ R.T
        u = ((rotated[:, 0] + 1) / 2 * (res - 1)).astype(int)
        v = ((rotated[:, 1] + 1) / 2 * (res - 1)).astype(int)
        sil = generate_silhouette(i, res)
        inside = sil[v, u].reshape(res, res, res)
        mask &= inside.astype(bool)

    return mask
```

2.4 Visualization

The carved voxel mask was visualized using 3D scatter plots in `Matplotlib`. This helped us qualitatively analyze how the dummy sphere forms through multiple-view carving.

3 Results

The following figures illustrate the output of our voxel-based reconstruction pipeline:



(a) Reconstructed 3D sphere voxel model

Figure 1: Voxel-based 3D model generation from synthetic silhouettes.

Observations:

- The carving successfully produced a roughly spherical 3D voxel structure.
- The density and accuracy depend on the number of views and voxel resolution.
- The dummy test validates that our carving logic functions correctly.

4 Discussion

This experiment confirmed that our voxel grid and silhouette-based carving method work effectively for simple shapes. The results demonstrate that the 3D volume can be refined progressively through multi-view projections. For real-world data, using actual silhouette images of diamond stones will produce the 3D volumetric representation of the object's geometry.

5 Conclusion

In this week's progress:

- Implemented dummy silhouette generation.
- Performed voxel carving to reconstruct a 3D sphere model.
- Validated voxel carving pipeline for further use on diamond silhouettes.

This marks a key milestone as we transition from grid visualization to volumetric object reconstruction. Our next step will involve applying this method to the diamond silhouette dataset to obtain the reconstructed 3D model.